Digital Video Fundamentals

- **Introduction to Video Compression**
- **From a Physical World to Digital Video**
- **Color Video**
- **Digital Video Format**
- **Why Is Video Compressible**
- **How Do We Evaluate Compression Performance**

Introduction to Video Compression

- **Why Do We Need Video Compression?**
  - Raw video requires an enormous amount of data.
  - Example: a 1920×1080 RGB video at 30 fps and 24 bits/pixel requires approximately.

$$1920 \times 1080 \times 24 \times 30 \approx 1.49 \text{ Gbit/s}.$$

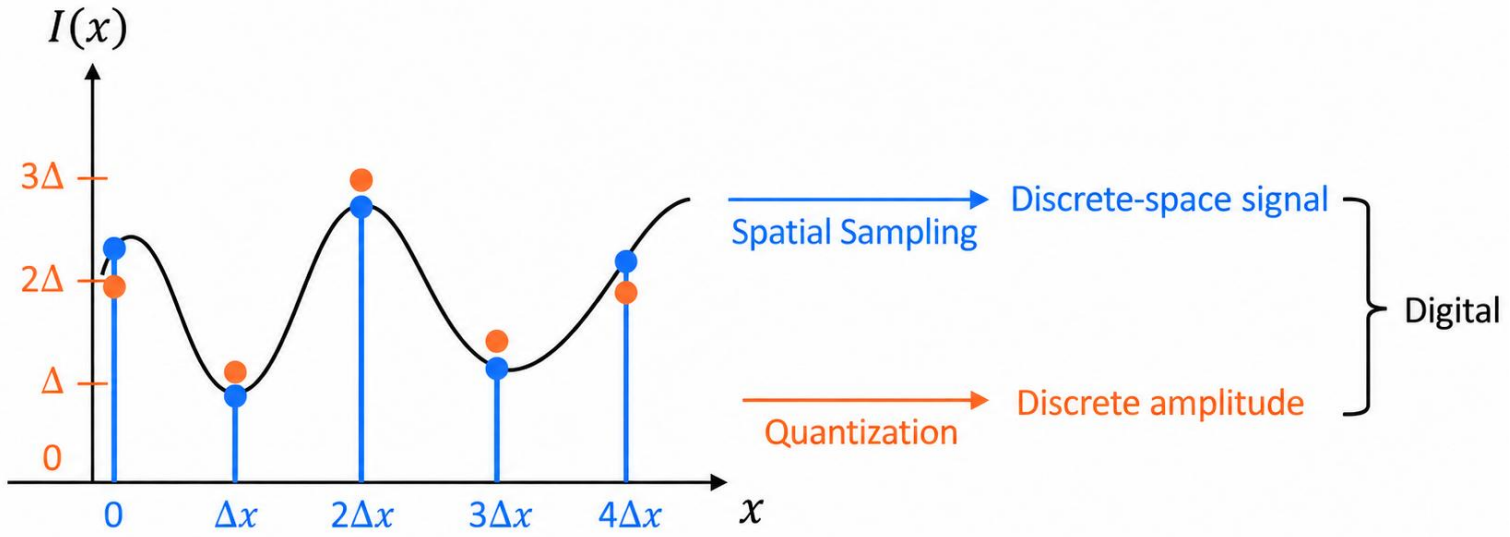
  - Such a data rate is impractical for storage and transmission.
  - Video compression significantly reduces the required bitrate while maintaining acceptable visual quality.
- **Applications**
  - Video streaming, Video conferencing, Digital broadcasting, Surveillance, Cloud gaming

From a Physical World to Digital Video

- **From Continuous to Discrete**
  - A real-world visual signal is continuous in space, time, and intensity.
    To represent it digitally, we need two fundamental operations:

    Sampling — discretize space and time (where and when to measure)

    Quantization — discretize signal values (what numerical value to store)
  - Continuous Scene → Sampling → Quantization → Digital Video



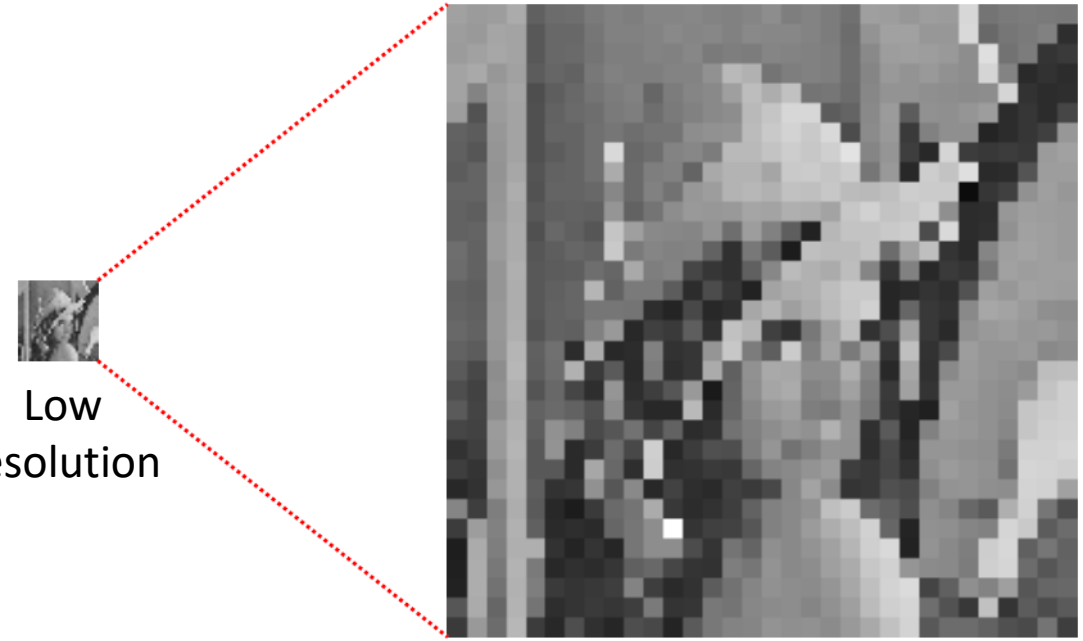
From a Physical World to Digital Video – Sampling

- **Spatial Sampling –** frame resolution
  - The continuous image is sampled along horizontal and vertical dimensions at discrete locations.
  - Higher sampling density → higher frame resolution (E.g., 256x256, 1920x1080, ..)



High resolution

Pixel : picture element



Low resolution

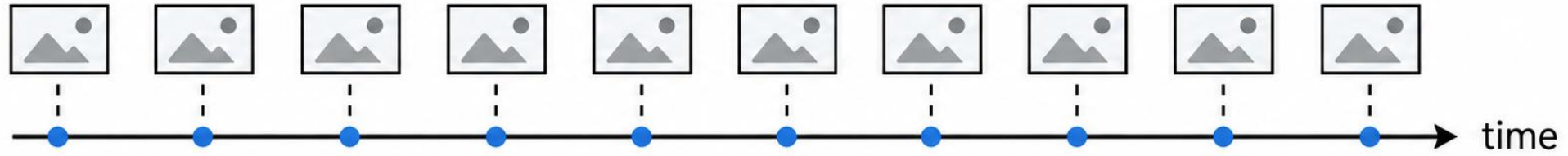
From a Physical World to Digital Video – Sampling

- **Temporal Sampling –** fame rate
  - Sampling the scene repeatedly over time. → Determine video **frame-rate**.
  - E.g., 24 fps, 30 fps, 60 fps, …

Low Sampling Rate (fewer frames per second)



High Sampling Rate (more frames per second)



From a Physical World to Digital Video – Quantization

- **Quantization –** bit-depth
  - Amplitude quantization determines the bit-depth of each pixel.
  - b bits → 2^b levels ➔ e.g., 8-bit → 256 levels, 10-bit → 1024 levels



8 bits per pixel
$2^8 = 256$



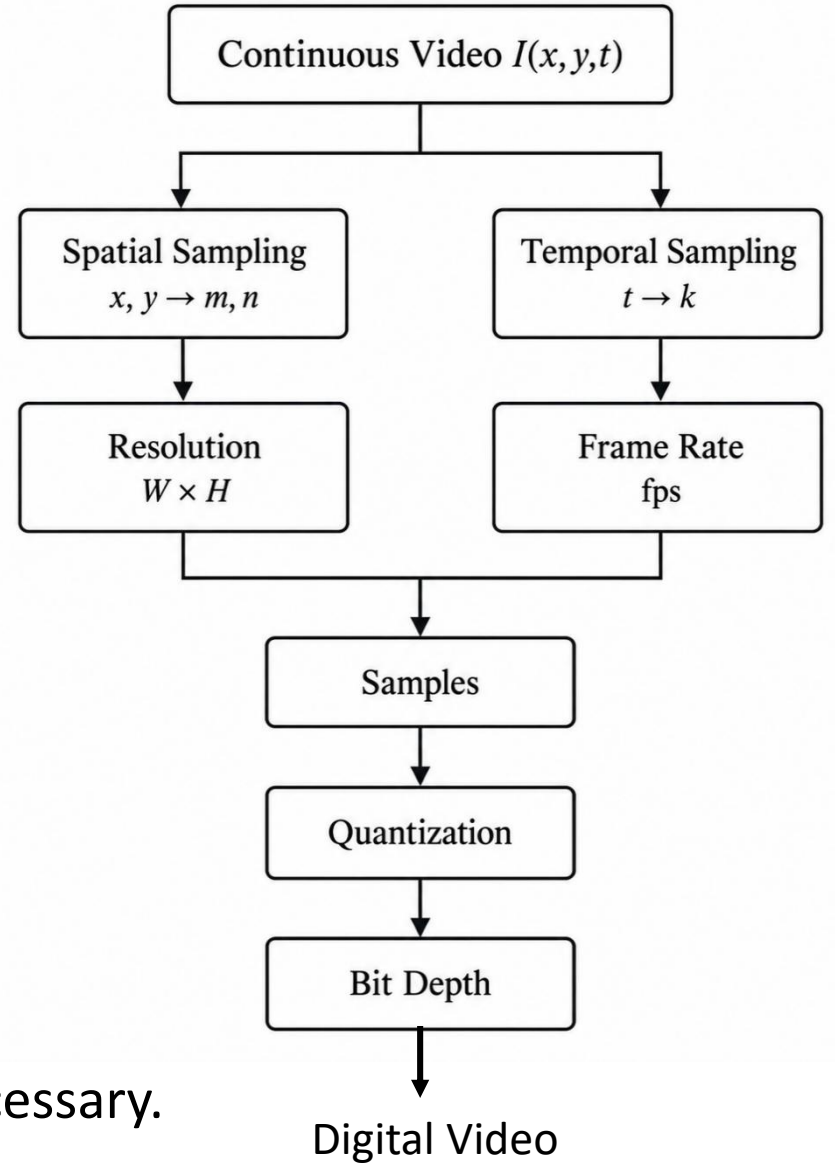
4 bits per pixel
$2^4 = 16$



2 bits per pixel
$2^2 = 4$

From a Physical World to Digital Video

- **Image Acquisition and Representation**
  - Spatial sampling determines resolution.
  - Temporal sampling determines frame rate.
  - Amplitude quantization determines bit depth.
- **Raw Video Data Rate**
  - R = W x H x fps x bits per pixel
- **Why compression is needed**
  - Higher resolution + higher frame rate + higher bit depth
  - → better representation quality
  - → much larger data rate
  - This enormous data rate is why video compression is necessary.



Color Video

- **Color Image Representation**
  - A color image is commonly represented using three primary color components: R, B, G (Red intensity, Green intensity, Blue intensity)
    - For an 8-bit RGB image → R,G,B ∈ [0,255]
  - Each pixel requires: 8 + 8 + 8 = 24 bits/pixel



1 bit per pixel

Black & white image



8 bits per pixel

Gray scaled image

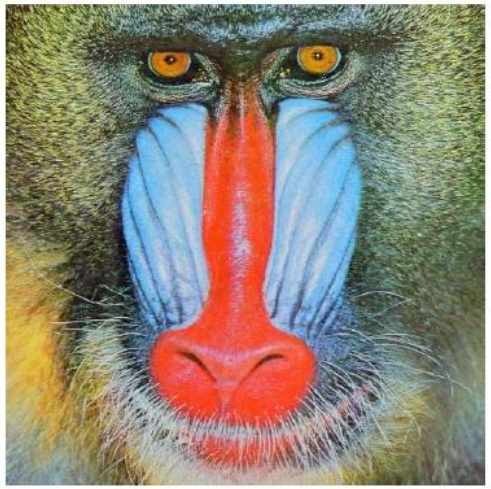


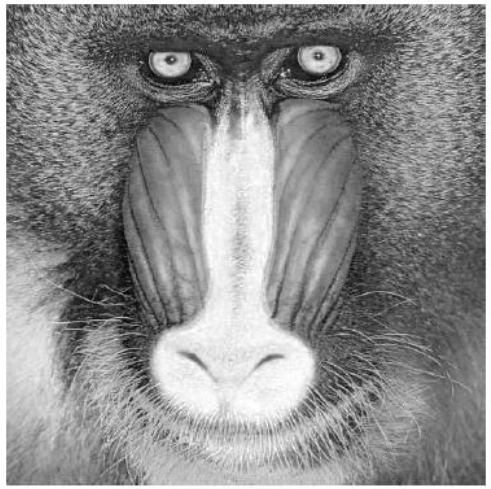
24 bits per pixel

Color image

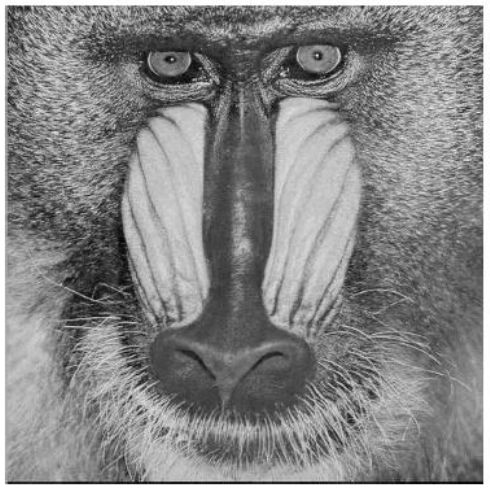
Color Video

- **RGB Color Image**

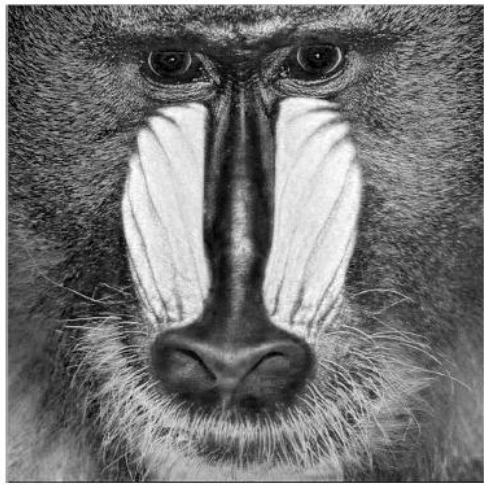




R



G



B

- A displayed RGB pixel contains three color values, but a camera sensor does not necessarily measure all three directly at the same location.
- How Does a Camera Capture Color?

Color Video

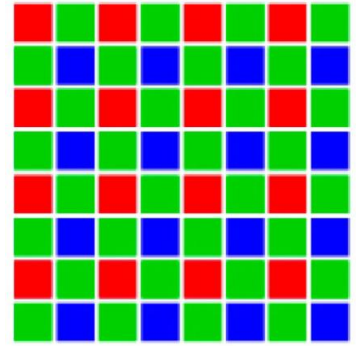
- **How Does a Camera Capture Color?**
  - There are two common approaches:

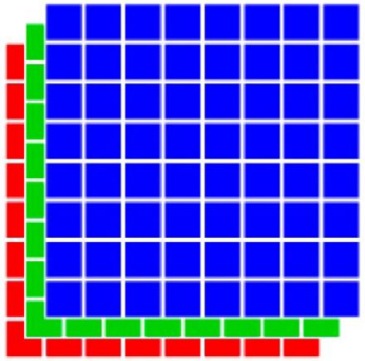
    1. Spatial Color Sampling

    - Single sensor + Color Filter Array (CFA)
    - The sensor measures different spatial locations for different color components.
    - Widely used in modern **CMOS and CCD** image sensors

    2. Depth-Based Color Sampling

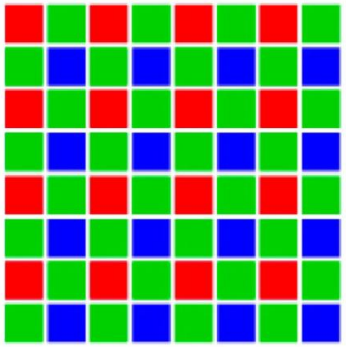
    - Foveon-like stacked sensor

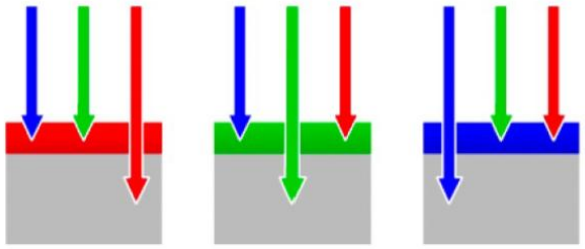




Color Video

- **Spatial Color Sampling**
  - Color Filter Array (CFA): Each photosite is covered by a color filter that makes it primarily sensitive to one color band.
    - Common arrangement: **Bayer pattern** (R: G: B = 25% : 50% : 25%)
    - Sensor measures only **one color component** at each location
  - Green is sampled more densely because human vision is more sensitive to luminance, to which green contributes strongly
  - The missing color components must be reconstructed by interpolation (de-mosaicing)

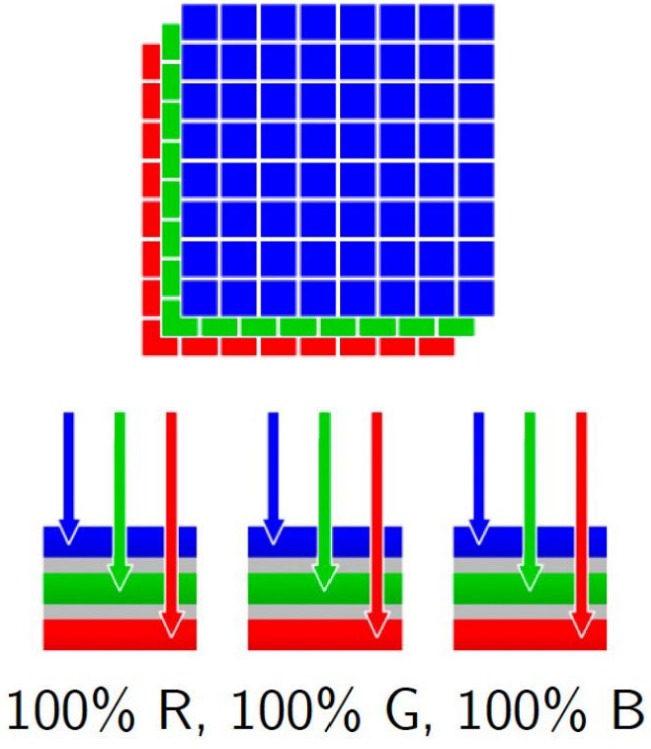




- **Advantages**
  - Simple single-sensor design
  - Low cost
  - High manufacturing efficiency → Widely adopted in cameras and mobile device

Color Video

- **Depth-Based Color Sampling**
  - Multiple light-sensitive sensor layers are stacked to capture different color information at the same spatial location
  - Different wavelengths penetrate silicon to different depths
    - B: shallow layer, G: middle layer, R: deep layer
  - Full RGB per pixel: Each spatial location captures information related to all three color components
- **Trade-offs**
  - More complex sensor structure
  - Color separation between layers may not be perfect - Noise and sensitivity can be more challenging
  - Less widely used than conventional Bayer-CFA sensors



Color Video

CMOS vs Foveon-like Sensor

Feature	CMOS (Bayer)	Foveon-like
Color Capture	One color per pixel	Full RGB per pixel
Sampling Method	Spatial	Depth-based
Demosaicing Needed	Yes	No
Manufacturing Cost	Low	High
Image Artifacts	More likely	Reduced
Color Resolution	Lower	Higher
Industry Adoption	Mainstream	Niche
Typical Usage	Smartphones, DSLRs	Specialized cameras

Digital Video Format

Category	Standard/Format	Resolution	Chroma Sampling	Frame Rate	Raw Data Rate*	Compression
Video Telephony	QCIF	176×144	4:2:0	30P	~9 Mbps	H.263 / H.264
Video conferencing	CIF	352×288	4:2:0	30P	~37 Mbps	H.261 / H.263
SD Internet Video	SIF	352×240 / 352×288	4:2:0	30P / 25P	~30 Mbps	MPEG-1
SDTV / DVD	BT.601 SD	720×480 / 720×576	4:2:0	60I / 50I	~124 Mbps	MPEG-2
Studio SD Video	BT.601 Studio	720×480 / 576	4:2:2	60I / 50I	~166 Mbps	MPEG-2 / DV
Professional SD	BT.601 Full	720×480 / 576	4:4:4	60I / 50I	~249 Mbps	Studio formats
HDTV 720p	SMPTE 296M	1280×720	4:2:0	24P / 30P / 60P	~265/332/664 Mbps	MPEG-2 / H.264
HDTV 1080i/p	SMPTE 295M	1920×1080	4:2:0	24P / 30P / 60I	~597 / 746 Mbps	MPEG-2 / H.264
Full HD Studio	1080p 4:2:2	1920×1080	4:2:2	60P	~1.49 Gbps	ProRes / DNxHD
Full HD RGB	1080p 4:4:4	1920×1080	4:4:4	60P	~2.24 Gbps	Cinema production
UHD 4K	UHDTV-1	3840×2160	4:2:0	60P	~5.97 Gbps	HEVC / AV1
UHD 4K Studio	4K 4:2:2	3840×2160	4:2:2	60P	~11.94 Gbps	ProRes / XAVC
DCI 4K Cinema	DCI 4K	4096×2160	4:4:4	24P / 60P	~12–24 Gbps	JPEG2000
UHD 8K	UHDTV-2	7680×4320	4:2:0	60P	~23.9 Gbps	HEVC / VVC
8K Studio	8K 4:2:2	7680×4320	4:2:2	60P	~47.8 Gbps	Broadcast production
8K RGB	8K 4:4:4	7680×4320	4:4:4	60P	~71.7 Gbps	High-end cinema

Digital Video Format

Resolution

Format	Typical Resolution
QCIF	176×144
CIF	352×288
SD	720×480
HD	1280×720
Full HD	1920×1080
UHD 4K	3840×2160
DCI 4K	4096×2160
UHD 8K	7680×4320

Chroma Sampling

Format	Color resolution
4:4:4	Full color
4:2:2	Half horizontal chroma
4:2:0	Half horizontal + half vertical chroma

Frame rate

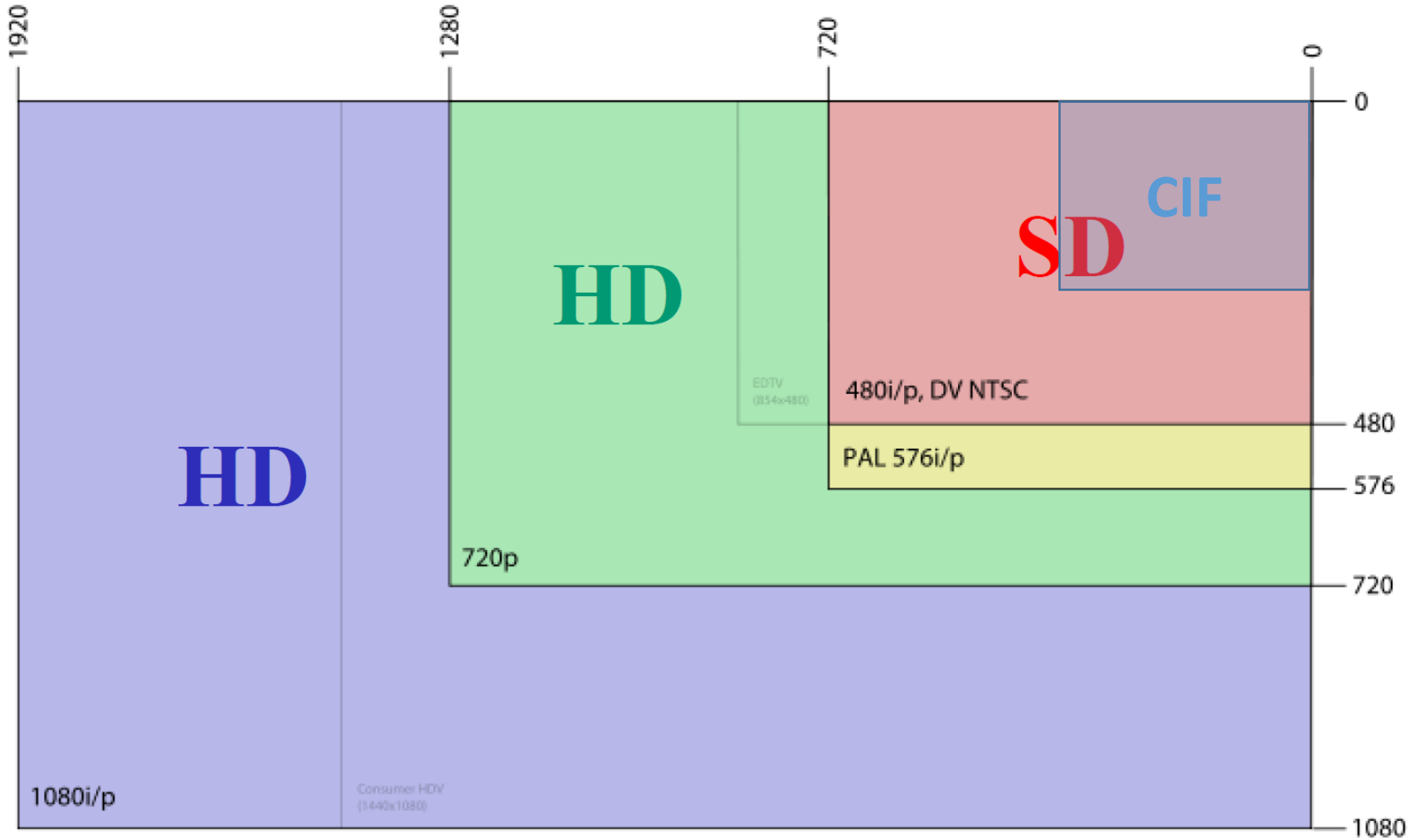
Frame Rate	Used in
24p	Cinema film
25p / 50i	PAL (50 Hz)
30p / 60i	NTSC (60 Hz)
50p / 60p	Modern digital video

Typical Codec

Era	Typical Codec
1990s	MPEG-1 / MPEG-2
2000s	H.264 / AVC
2010s	HEVC / VP9
2020s	VVC / AV1

Resolution

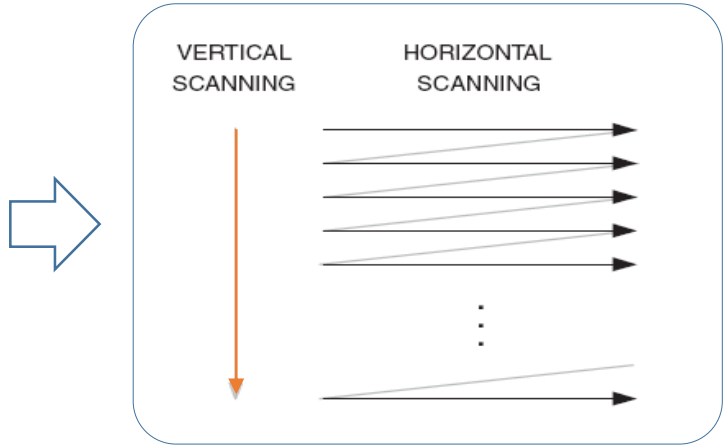
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SD	720×480
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Progressive vs Interlace

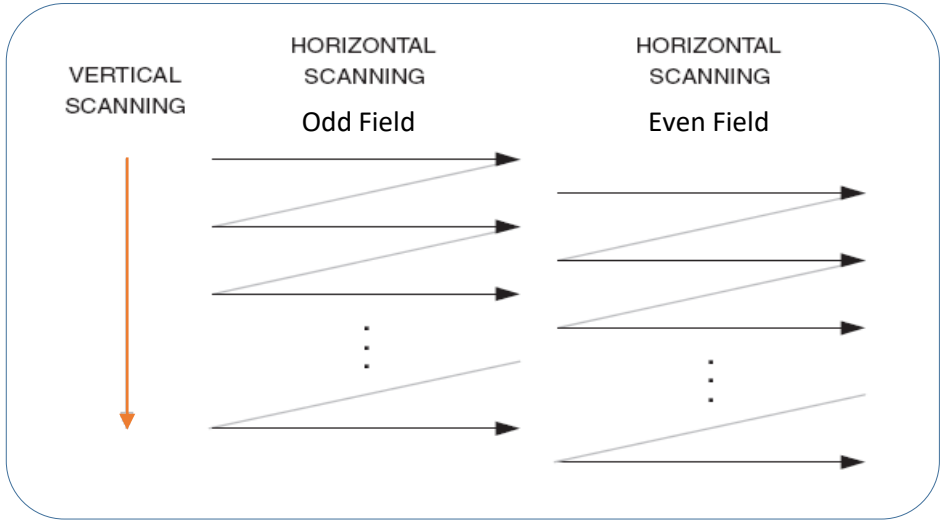
Progressive Video (P)

- Each image contains all scan lines.
- One complete image is called a **frame**.
- Provides full spatial resolution in every frame.
- Common in modern displays and streaming systems.



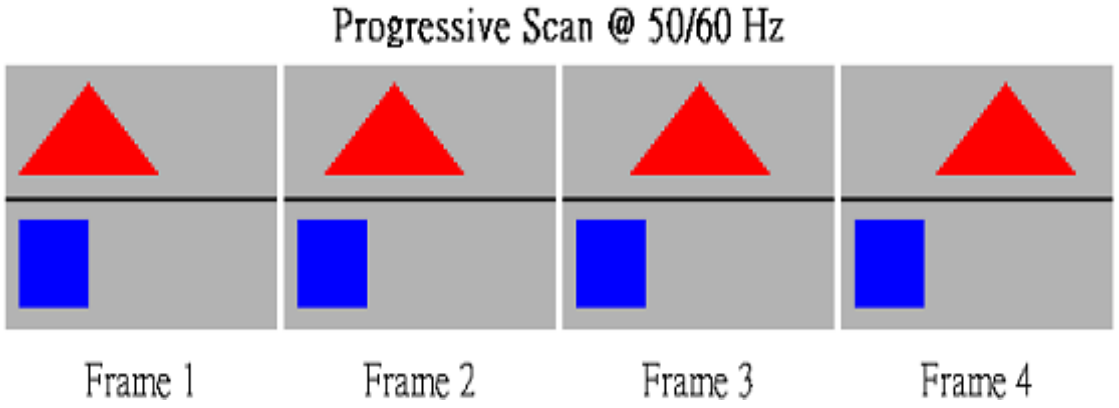
Interlaced Video (I)

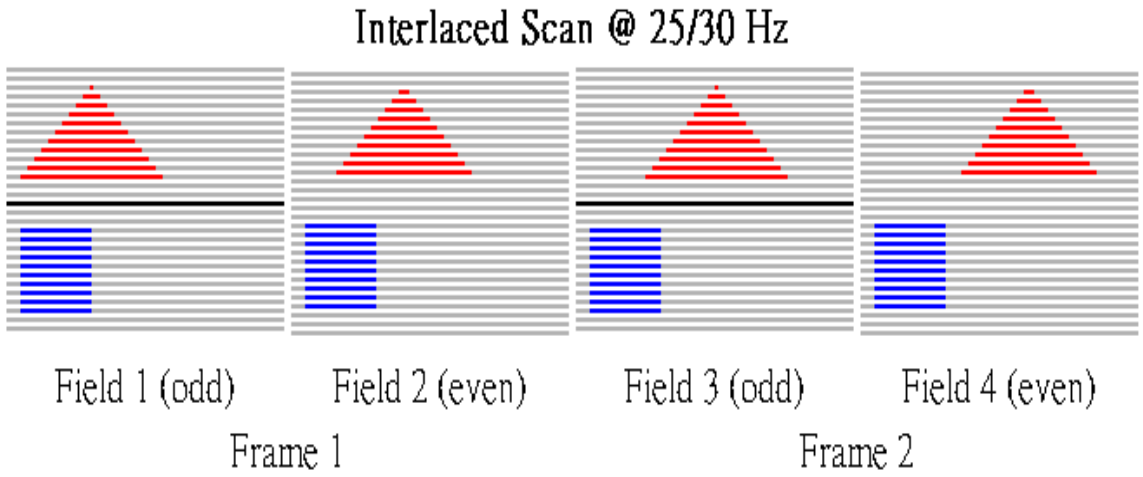
- Each image contains only half of the scan lines.
- One partial image is called a **field**.
- Two field types:
  - **Odd field:** lines 1, 3, 5, ...
  - **Even field:** lines 2, 4, 6, ...
- One odd field + one even field form a **frame**.
- Why does it work?
  - Odd/even fields are displayed rapidly in succession.
  - Due to persistence of vision, they are perceived as a continuous image.



Progressive vs Interlace

- **Why Was Interlace Introduced?**
  - In analog TV broadcasting system, the channel bandwidth required for transmitting full scan lines is too high. → Interlaced video
- **Drawbacks of Interlace video**
  - Differences between adjacent fields can produce interline **flicker**
  - Reduces compression efficiency compared to progressive video.



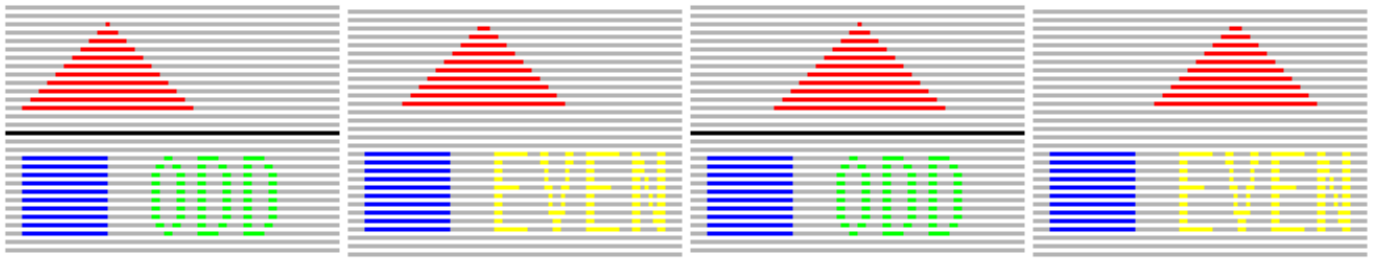


Progressive vs Interlace

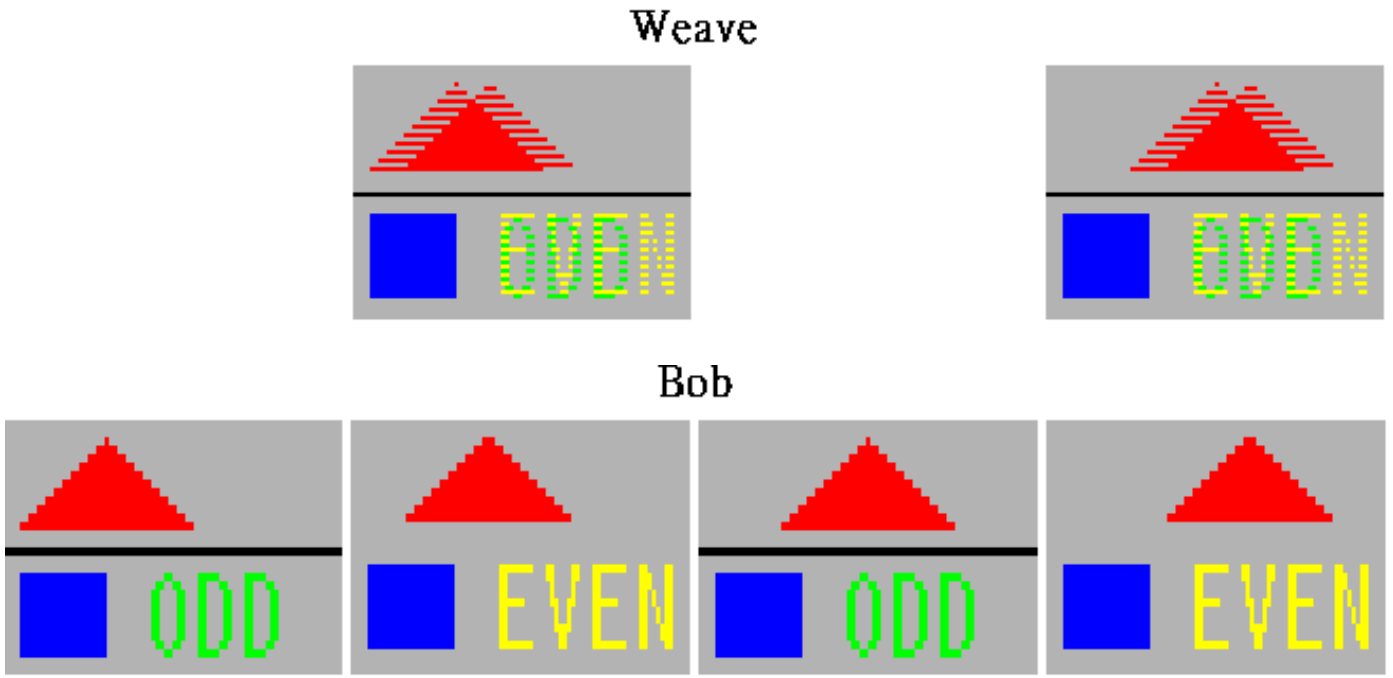
- **De-Interlacing**
  - Modern LCD and plasma displays are inherently progressive.
  - Interlaced video must be converted before display. → called **de-interlacing**
- **Basic De-Interlacing Methods**
  - **Weave** - Combines two consecutive fields into one frame.
    - Preserves full vertical resolution.
    - May produce combing/feathering artifacts in moving regions.
  - **Bob** - Generates a frame from a single field using line doubling
    - Handles motion better than weave.
    - May introduce flicker or reduced vertical detail.
- **Advanced De-Interlacing**
  - More sophisticated algorithms can improve visual quality.
  - Better de-interlacing usually requires higher computational complexity.

Progressive vs Interlace

Interlaced video



De-interlacing



Frame Rate

- To reduce flicker and electrical interference, frame rate is synchronize with power frequency
- **PAL Regions (50 Hz power)**
  - TV refresh synchronized with 50 Hz electrical systems
  - Common formats: **50i** = 50 fields/sec = 25 frames/sec
    25p = 25 progressive frames/sec
- **NTSC Regions (60 Hz power)**
  - TV refresh synchronized with 60 Hz electrical systems
  - Common formats: **60i** = 60 fields/sec = 30 frames/sec
    30p = 30 progressive frames/sec

Frame rate

Frame Rate	Used in
24p	Cinema film
25p / 50i	PAL (50 Hz)
30p / 60i	NTSC (60 Hz)
50p / 60p	Modern digital video

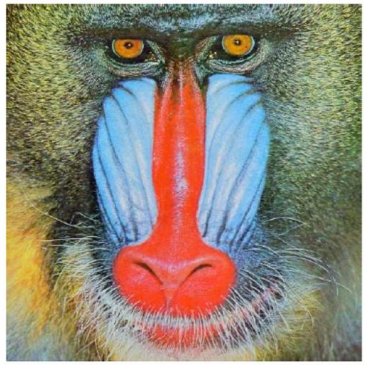
- **Why does cinema use 24p?**
  - **24 fps** became the minimum frame rate for smooth motion while minimizing film consumption.
  - Originated from film cameras and movie projectors. Still widely used in movie, streaming films

Color Space

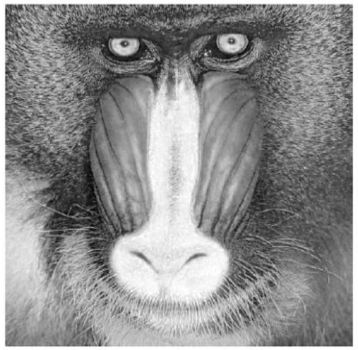
- Mixture of luminance and chrominance
  - RGB → display
  - CMY → printer
- Separation of luminance and chrominance
  - YUV – PAL (analog TV system)
  - YIQ – NTSC (analog TV system)
  - YCbCr - digital video (used in JPEG, MPEG, H.264, HEVC, AV1)
  - HSI - hue, saturation, intensity (brightness) I = (R+G+B)/3
  - HSV – hue, saturation, value (brightness) V = max (R,G,B)
- Color space conversion – better representation for processing or data reduction

Color Space

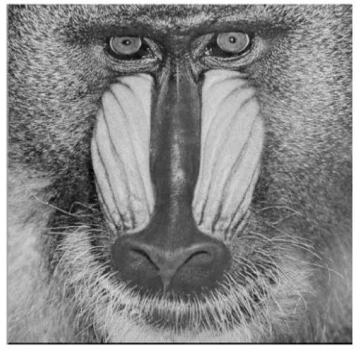
- **RGB, YUV, YIQ, YCbCr**



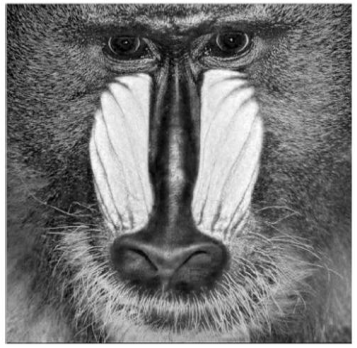
Original color image



R



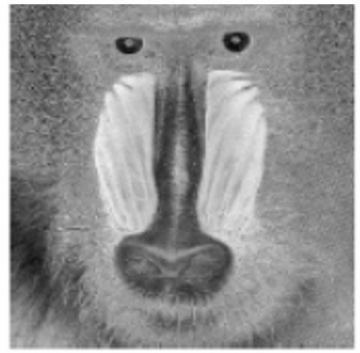
G



B



Y



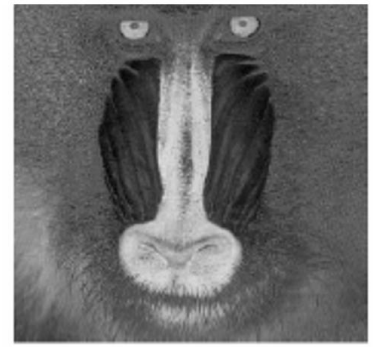
U



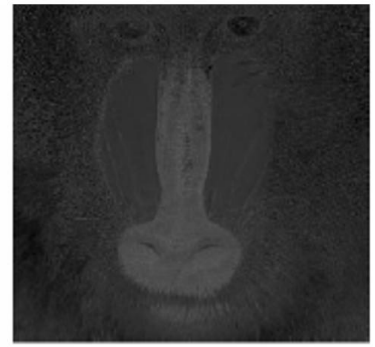
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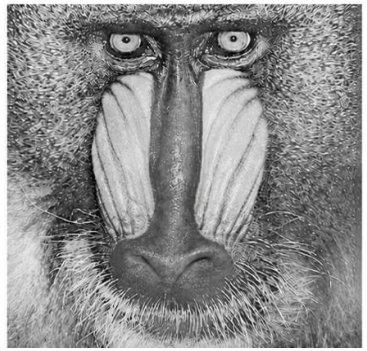
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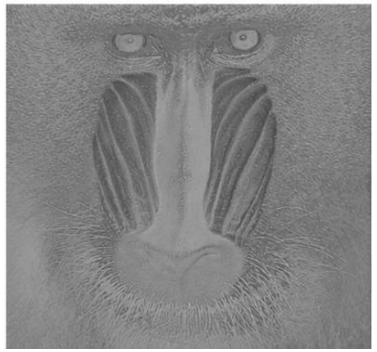
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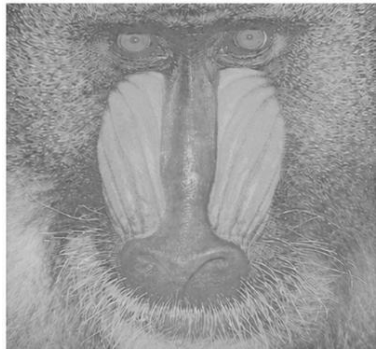
Q



Y



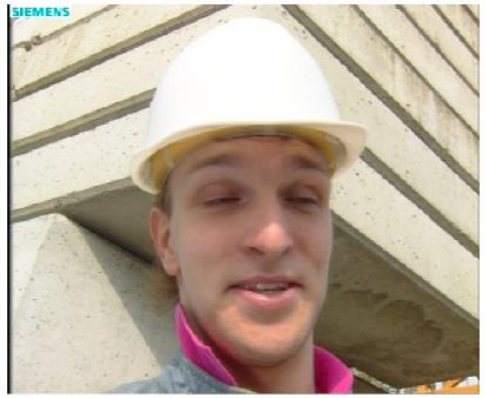
Cb



Cr

Color Space

- **RGB vs YCbCr**



Original



R



G



B



Y



Cb



Cr

Color Space Conversion

- **Color Space Conversion**

  (R, G, B) → (Y, U, V)

  $$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

  - Y: Brightness
  - U = 0.492(B-Y), V=0.877(R-Y): color differences
  - Brightness contribution (G > R > B)

    Y = 0.299R + 0.587G + 0.114B

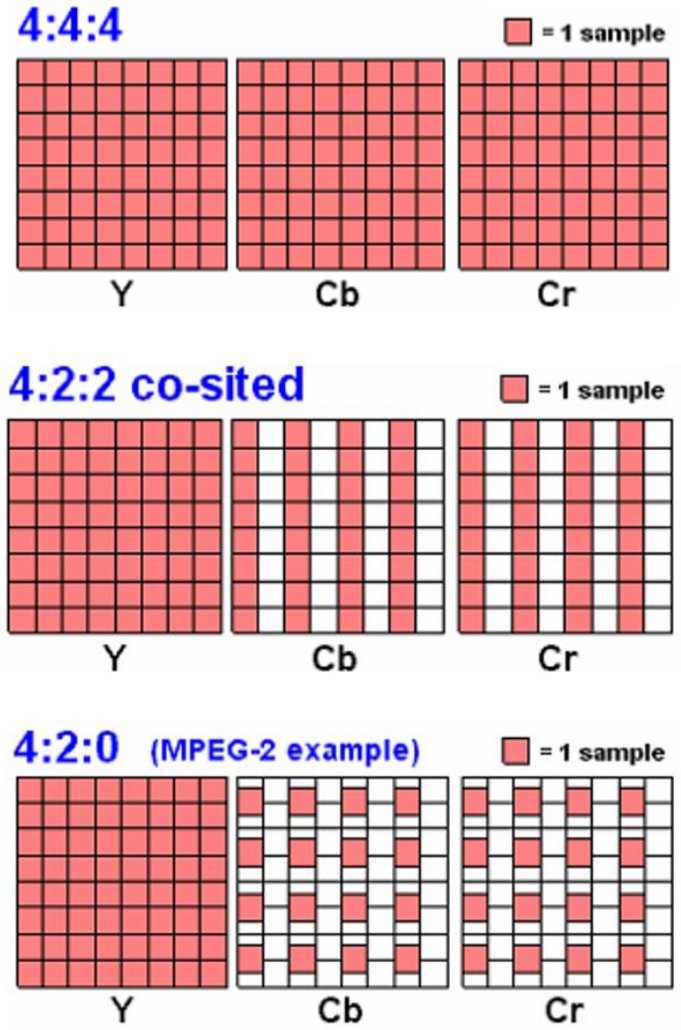
    ← Human vision is more sensitive to **green** than red and blue.
  - Coefficients of the first row add up to 1, those of the other rows to 0
    - R = G = B → gray image (U = V = 0)

  (R, G, B) → (Y, Cb, Cr)

  $$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.257 & 0.504 & 0.098 \\ -0.148 & -0.291 & 0.439 \\ 0.439 & -0.368 & -0.071 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 16 \\ 128 \\ 128 \end{bmatrix}$$

Chroma Sub-sampling

- **Chroma Sub-sampling on YCbCr**
  - Use low resolution for chrominance because eye is more sensitive to grayscale than to color.
  - Chroma (Cb and Cr) are sampled at a lower rate than Luma (Y).
  - 4:4:4 – Cb and Cr are sampled at the same rate as Y
  - 4:2:2 – Cb and Cr are sampled at ½ resolution of Y
  - 4:2:0 – Cb and Cr are sampled at 1/4 resolution of Y



Why Is Video Compressible?

- **Video contains a large amount of redundancy**

Spatial Redundancy

Example: 120 121 121 121
120 122 122 121

- Neighboring pixels in the same frame are often similar.
- Instead of coding every pixel independently, we can predict pixels from nearby reconstructed samples.

Temporal Redundancy

- Consecutive frames are usually similar. A moving object may simply change its position from one frame to the next.
- Instead of coding the entire new frame, we can describe: where the object moved, and the remaining prediction error

Statistical Redundancy

Example: 0, 0, 0, 1, 0, 0, −1, 0,…

- Some symbols occur much more frequently than others.
- More frequent symbols can be represented using fewer bits.

Why Is Video Compressible?

- **Perceptual Irrelevance**
  - Video contains a large amount of perceptually less important information.
  - The human visual system is less sensitive to some visual information than to others.
  - Such information can be represented with lower precision or partially discarded with little noticeable loss in visual quality.

- **Video compression exploits signal redundancy and perceptual irrelevance.**
  - Spatial → intra prediction / transform
  - Temporal → inter prediction / motion compensation
  - Statistical → entropy coding
  - Perceptual → chroma subsampling / quantization / perceptual optimization

Lossless vs. Lossy Compression

- **Lossless Compression**
  - The decoded signal is identical to the original: $\hat{X} = X$
  - No information is lost.
  - Exploits **spatial, temporal, and statistical redundancy** without discarding information.
  - Usually provides a limited compression ratio.
  - Examples: ZIP, PNG
- **Lossy Compression**
  - The decoded signal is not exactly identical: $\hat{X} \neq X$
  - Exploits spatial, temporal, and statistical redundancy, and additionally discard **perceptually less important information**.
  - Lossy compression achieves much higher compression ratios.
  - Commonly used in video codec

How Do We Evaluate Compression Performance

- **A good video codec should provide:**
  - low bitrate,
  - high visual quality,
  - reasonable encoding and decoding complexity.
- **Bitrate**
  - Measured in: kbps, Mbps, Gbps
  - Compression Ratio (CR): $\mathrm{CR} = \frac{\text{Uncompressed Size}}{\text{Compressed Size}}$
- **Quality Metrics**
  - Objective Quality Metrics
  - Subjective Quality Evaluation

Objective Quality Measure

- **MSE** (Mean Squared Error)

$$MSE = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [I(x,y) - K(x,y)]^2$$

  - *I*(x, y): original pixel, K(x, y): degraded pixel, MxN: image size

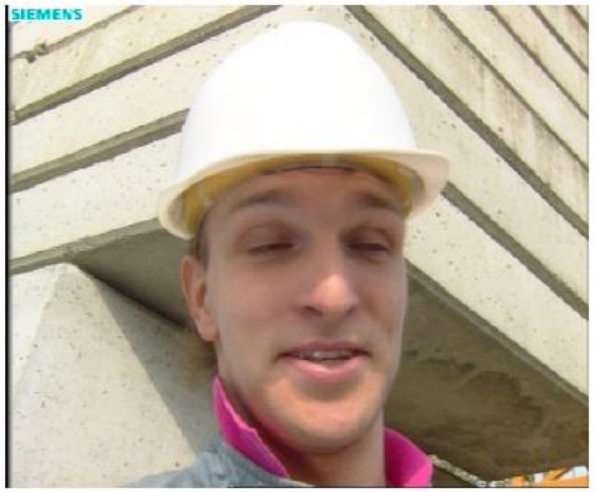
- **PSNR** (Peak Signal-to-Noise Ratio)

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

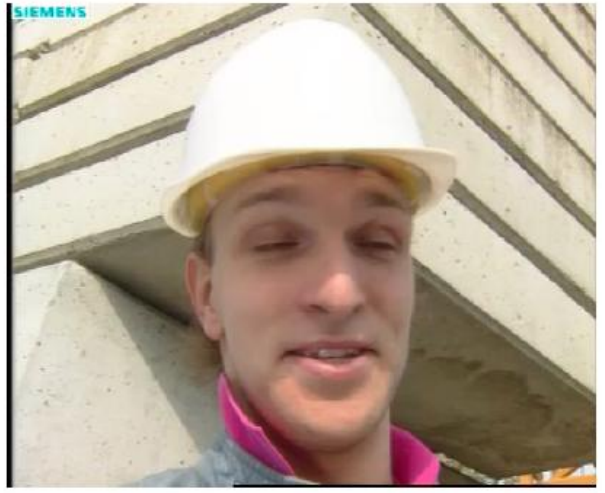
  - MAX_I : peak pixel value (255 for 8 bits, 1023 for 10 bits)
  - Higher PSNR generally indicates lower reconstruction error.

Objective Quality Measure

- **PSNR**
  - Excellent (>40)
  - Good (30-40)
  - Poor (<30)



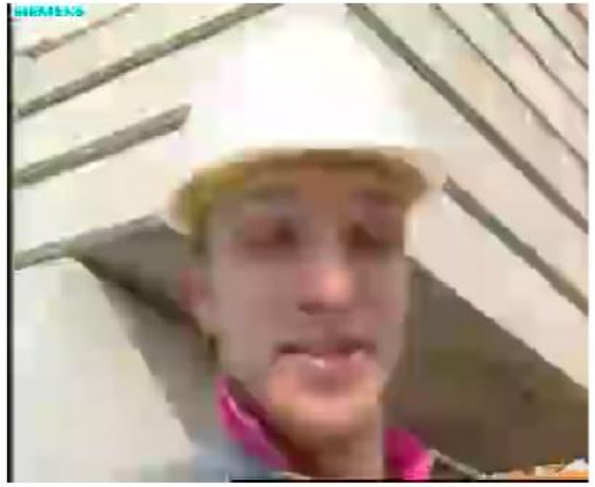
Original



42.6



35.5



28.7

Objective Quality

- Both PSNR & MSE do not correlate well with human perception

MSE = 27.10



MSE = 21.26



Subjective Quality Evaluation

- **Human Visual System (HVS)-based evaluation**
  - **MOS (Mean Opinion Score)**
    - Human observers' judgement of visual quality
  - **Methods defined in ITU-R BT.500-11**

    1. **DSCQS** (Double Stimulus Continuous Quality Scale
       - Reference and degraded video graded in a randomized order. (reference unknown)
    2. **DSIS** (Double Stimulus Impairment Scale)
       - Reference first and then the degraded video
       - Grade the degraded video relative to its reference. (reference is known)
    3. **SSCQE** (Single Stimulus Continuous Quality Evaluation)
       - Grade the degraded video in a continuous manner without a reference